Chapter 11 exercises • File-system interface

Build an explicit state/variable table before answering. Worked answers are distributed separately.

OS11-E01 • Symlinks followed by mount crossings

All directories are searchable. `/a/link` targets `../b`; F7 is mounted at `/b`. Directory `/x/y` and object `/x/z` exist, and `/a/other` targets `/x/y`.

1. Which object does `/a/link` finally resolve to?
2. Resolve `/a/other/../z` and contrast cancellation of the path string before resolution.
3. Distinguish hard-link and symlink identity and cross-filesystem behavior. How can TOCTOU substitution be controlled?

OS11-E02 • Actual OFD offsets

Data is ABCDEFGHIJ. P.fd3=open(F)→O1 and P.fd4=dup(fd3). Q.fd8 independently opens F→O2. P then forks C.

1. In order, P reads three bytes through fd3, C reads two through fd4, and Q reads four through fd8. Give the data and both offsets after each step.
2. P calls pread(fd3,2,1), then read(fd4,2). Give both results.
3. If P/C concurrently use lseek+read, what does pread solve, and what does it leave unsolved?

OS11-E03 • Named users and the mask

Owner is Alice and file group dev. ACL is user::rw-, user:bob:-w-, group::r--, mask::r--, other::---. Bob is also in dev, without additional privilege.

1. What may Alice, Bob, and an ordinary dev member do?
2. If the mask changes to rw, what are Bob's read/write rights and the displayed mode group class?
3. Bob opened O_WRONLY while mask=rw; the mask then becomes r. Compare the existing descriptor with a new open.

OS11-E04 • Record positions and index I/O

Fixed 32 B records occupy 1024 B blocks. A thousand records are sorted by unique keys 0…999. Each block's first-key index entry occupies 16 B, and the index is in RAM. Use binary upper-bound search on the index for the greatest first key not exceeding the target. Count only index-key comparisons; data-block search is separate.

1. Give record 77's byte position, file block, displacement, and final-block waste.
2. Give index size, candidate block, worst-case comparisons, and data I/O count.
3. Why does jL fail for variable-length records?

OS11-E05 • Short writes and zero progress

Write 6000 B from initial offset 100, with successful results 2500,1500,2000. In another scenario the second call returns zero.

1. Give each buffer start, remaining count, and resulting offset.
2. State the general loop invariant and the premise for termination.
3. Distinguish handling of zero, EINTR, and EAGAIN.

OS11-E06 • An unnamed but live object

F0:42 has hard links /a,/b. O1 is the only live reference, is open for reading and writing, and has offset 7; there are no other open, mmap, or other live references. Let fd refer to O1. Successfully unlink a, unlink b, ftruncate(fd,0), then close(fd). For part 3 separately assume tmp and target are different objects on the same mounted filesystem and that rename succeeds with required permissions and compatible types.

1. Give link counts and reclamation eligibility at each step.
2. After truncate, give the offset and read result.
3. If rename replaces target with tmp, does an old descriptor follow the name to the new inode?

OS11-E07 • Mount hiding and permissions

F0's /mnt contains old; F7's root contains new and has no old. P opens old before F7 is mounted at /mnt. All ancestor directories are searchable. Separately D has mode0711 and its file F mode0644; caller is other. Exclude read-only mounts, immutable flags, MAC and other extra restrictions.

1. After mount, evaluate /mnt/old, /mnt/new, and P's existing descriptor.
2. Can other list D or read F through a known name?
3. If F has mode000 and parent D mode0733, can other unlink known F? Exclude sticky and privilege rules.

OS11-E08 • Two groups and a combined request

Carol belongs to dev and qa; the file owning group is dev. Its ACL is user::rw-, group::r--, group:qa:-w-, mask::rw-, other::rwx. Someone else owns the file; Carol has no named-user entry, additional privilege, or other restrictions. Read+write is one combined Linux POSIX ACL access check.

1. Evaluate read, write, and one combined read+write request.
2. What changes if mask becomes r?
3. If user:carol:--- is added, can other authorize Carol?

OS11-E09 • Visibility, exclusion, and persistence

Initially value 0. A/B open; A writes 1; B reads; A closes; B reads in its existing session; newly opened C reads. Every read addresses the same data position to compare versions, rather than inheriting the previous read's advanced offset. Also publish a new version using tmp. Assume the same mounted filesystem supports fsync on both regular files and parent directories, the device honors successful flush durability, every operation succeeds, and no other writer intervenes.

1. Compare the three read values under an immediate shared image and session snapshots.
2. Can advisory locking stop an uncooperative writer? Do [0,100) and [100,200) conflict?
3. Give the required steps and assumptions for a durable published name.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 11. Linux open(2); rename(2); acl(5); path_resolution(7); credentials(7).

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https://github.com/xinhi6716/computer-science-self-study